KEEP EXPLORING

BUILD.
HACK.
PLAY.

Discover the hardware design, game code, wiring guides and 3D-printable console and gamepad models on GitHub.

SCAN FOR THE PROJECT

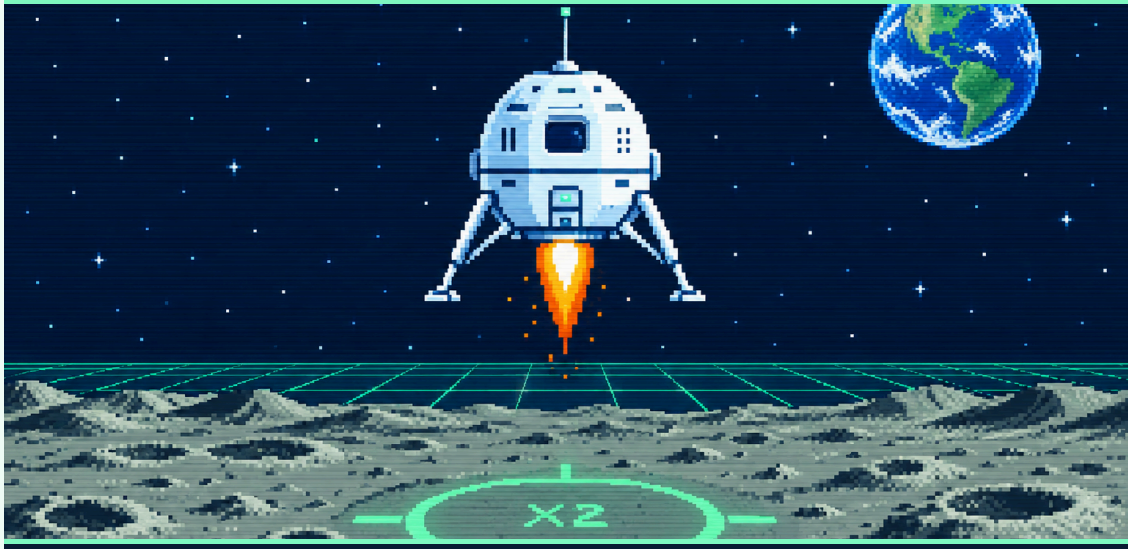
github.com/melrouts/RetroConsole-LunaStation

Luna Station is a hobby project by Eduard Garanskij.
Created for Retro Computer Festival 2026 in Cambridge.
Contact: retro@lea.lt

LUNA 9
1966 — 2026

LUNA STATION

A retro console built from logic up



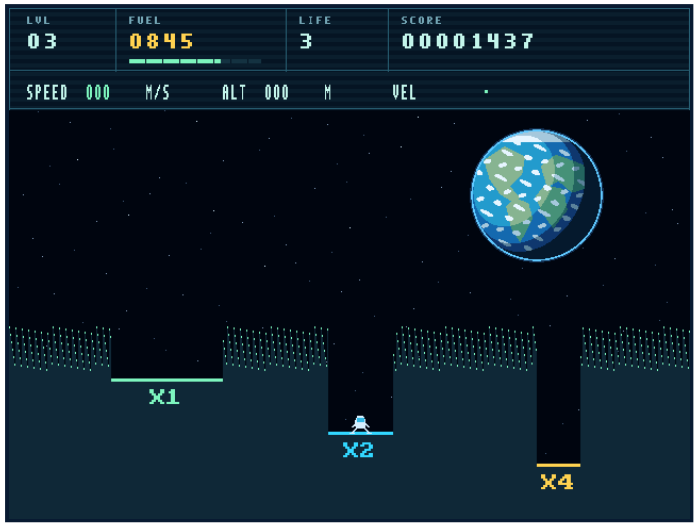
A hobby project celebrating **60 years since Luna 9's first soft landing on the Moon**.

Play Lunar Lander on a homemade FPGA console.
Retro Computer Festival · Cambridge · 2026

01 / THE GAME

LAND IT IF YOU CAN

Gravity never stops. Rotate your lander, use just enough thrust and touch down gently before your fuel runs out. Narrower pads pay bigger bonuses — if you can land safely.



Actual gameplay, generated by the FPGA.

GAMEPAD CONTROLS

LEFT / RIGHT	Rotate the lander
THRUST	Start the game or fire the engine
MENU	Go back or open the menu

> With the optional keyboard bridge: **A / D** rotate, **W** thrusts and **B** opens the menu.

02 / INSIDE THE CONSOLE

A COMPUTER FROM SCRATCH

An FPGA is a chip whose logic can be configured to become a custom computer. Luna Station combines that hardware with a tiny game program.

BOARD	Sipeed Tang Nano 9K FPGA
CPU	32-bit PicoRV32 RISC-V, 27 MHz; 8 KiB application RAM
VIDEO	HDMI, 640 × 480 at 60 Hz
GAMES	Booted from a FAT-formatted SD card
SOUND	ESP32 synthesizer and MAX98357A amplifier
INPUT	ESP-NOW wireless gamepads; optional LED matrix



Console case design



Gamepad case design

Case images are CAD renders. The project is a work in progress: check the wiring and printed parts for your own hardware.